

Hackathon Mastery & Learning Compendium

Curated Resource Vault, Autonomous Learning Freedom, Jury Defense Strategy, and Subsystem Deliverables

EVENT

Smart Horizon 48H Hackathon

GRAND FINALE

Sept 3–5, 2026 @ NHCE BLR

TEAM STRATEGY

Full Study Autonomy

CORE DIRECTIVE

Deliverable Mastery & Defense

💡 Autonomous Learning Freedom: Recommended & Normalized Reference Blueprint

To All Team Members: All resources, recommended hours, and topic suggestions in this document are **completely optional reference baselines**. As teammates have already been proactively self-learning over time, you are **100% encouraged and empowered to follow your own preferred courses, documentation, or study rhythm**. There is no rigid daily timetable. The only shared mission is satisfying your **subsystem deliverables and conceptual mastery** so that our team can effortlessly explain, code, and defend our architecture live on stage.

1 The Hackathon Reality: Why First-Principles Defense is Essential

⚠️ Official Rule Alert: Strict Scrutiny on Copy-Pasted & Black-Box AI Code

Smart Horizon Rules (Sections 5.2, 5.3, 6.1 & 6.2): All solutions must be developed live during the 48-hour event. *"Reuse of previously developed projects, codebases, or prototypes is not allowed... Plagiarism will result in immediate disqualification."*

🎯 The "Jury Viva" Reality: Defending AI-Assisted Work Confidently

AI assistants are incredible tools during preparation, but **the jury will aggressively interrogate each teammate during live viva evaluations**. Judges will inspect Git commit trails, test line-by-line understanding, and ask team members to adjust parameters or rewrite algorithms live on stage.

1. LINE-BY-LINE FLUENCY

Understand every import, ROS2 callback, tensor operation, and packet structure in your module.

2. LIVE CODE ADAPTATIONS

When a judge asks to modify a timeout, add a geofence, or alter a setpoint rate, execute it live.

3. EMPIRICAL INTEGRITY

Present live captured terminal benchmarks (pytest stdout, Gazebo RTF, topic Hz) rather than synthetic mocks.

2 Project SUTRA Architecture & Subsystem Assignments

Project SUTRA (Swarm Unified Tactical Reconnaissance Architecture) coordinates autonomous drone swarms for GPS-denied search and rescue. Each teammate leads one core subsystem:

SUBSYSTEM	LEAD MEMBER	CORE TECH STACK	PRIMARY MISSION OBJECTIVE	KEY SYSTEM INTERFACE
Subsystem A: GNC & Control	Nikhil (Lead)	PX4 Autopilot, ROS2, ORCA 3D, VIO	50Hz trajectory control, GPS-denied avoidance	/sutra/gnc/{id}/pose_stamped
Subsystem B: Comms & Sim	Nikhil (Lead)	802.11s Mesh, SwarmRAFT, Gazebo Sim 8	<100ms consensus failover, digital twin SITL	/sutra/comms/mesh_heartbeat
Subsystem C: AI Perception	Vedanth Sai Ram	YOLOv8-Nano, TensorRT, PyTorch, WGS84	Edge survivor/threat detection & raycasting	/sutra/perception/targets
Subsystem D: 3D GIS GCS	Siva Kesava	Flask, WebSockets, Leaflet.js, REST APIs	Live tactical telemetry HUD & 1-click RTL	ws://0.0.0.0:9090
Subsystem E: Verification	Harika	pytest, CI/CD, DOCS.md, Deck Design	G1–G6 verification gates & empirical audit	pytest sutra_ws/src/*/test/
Subsystem F: Tactical Ops	Rohith Kumar	NDMA CONOPS, SOPs, Kedarnath Case Study	Disaster scenario profiles & rescue defense	docs/conops/DOCS.md

3 Curated Domain Resource Vault (Part 1: Core Systems)

Choose resources that match your personal learning style (video walkthroughs, structured courses, or official API documentation).

1. Python Fundamentals & Core Software Engineering

UNIVERSAL SKILL

Crucial for writing clean ROS2 nodes, test fixtures, Flask routes, and AI inference scripts without syntax friction.

RESOURCE NAME	TYPE / CREATOR	DURATION	FOCUS / STRENGTHS	LINK
Python Tutorials for Beginners	YouTube (Dave Gray)	7 hrs (72 vids)	Clear, beginner-to-intermediate pacing & practical tasks	Watch Playlist ↗
Python Tutorials 2024	YouTube (Indently)	12 hrs (72 vids)	Modern Python 3.12 features, typing, OOP, and patterns	Watch Playlist ↗
CS50's Intro to Programming with Python	Harvard University (edX)	Self-Paced	Gold standard university curriculum (functions, OOP, regex)	Enroll Free ↗
Python for Everybody	Univ. of Michigan (Coursera)	Self-Paced	Data structures, JSON handling, REST API integration	Enroll Free ↗
Automate the Boring Stuff with Python	Al Sweigart (Free Book)	Read / Practice	Practical scripting, file I/O, parsing, and automation	Read Book ↗

2. ROS 2 Robotics Middleware (Humble / Jazzy)

NIKHIL, VEDANTH, SIVA

The backbone of SUTRA inter-node communication, pub/sub QoS profiles, services, actions, and multicasting.

RESOURCE NAME	TYPE / CREATOR	DURATION	FOCUS / STRENGTHS	LINK
ROS 2 for Beginners - 2025	YouTube (The Construct)	8 hrs (10 vids)	Hands-on guide to ROS2 workspaces, nodes, and QoS	Watch Playlist ↗
Learn ROS2 Jazzy Crash Course 2025	YouTube (The Construct)	4 hrs (1 vid)	Fast-paced complete overview of ROS2 concepts and tooling	Watch Video ↗
Official ROS 2 Humble Documentation	Open Robotics	Docs / Tutorials	Definitive reference for custom interfaces, launch files, rclpy	Read Docs ↗
Robotic Manipulation (MIT 6.800)	MIT OpenCourseWare	Lecture Series	Deep mathematical foundation of spatial transforms & control	Explore Course ↗

3. Gazebo Sim 8 Digital Twin & PX4 Drone Autopilot

NIKHIL (LEAD), VEDANTH

Essential for building disaster digital twins, multi-drone physics simulation, and offboard MAVLink/MAVSDK control.

RESOURCE NAME	TYPE / CREATOR	DURATION	FOCUS / STRENGTHS	LINK
Gazebo Simulator Tutorials	YouTube (Robotogeddon)	4 hrs (9 vids)	SDF model creation, sensor plugins, world physics configuration	Watch Playlist ↗
Official Gazebo Sim Tutorials	Gazebo Sim (Official)	3 hrs (8 vids)	Modern Gazebo Garden/Harmonic bridges to ROS 2	Watch Playlist ↗
PX4 Drone Programming Series	YouTube (RIIS LLC)	8 hrs (22 vids)	MAVLink, offboard position setpoints, SITL setup	Watch Playlist ↗
PX4 Autopilot Developer Guide	PX4 Community	Docs	Offboard mode, uORB architecture, ROS 2 micro-XRCE-DDS bridge	Read Docs ↗

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Curated Domain Resource Vault (Part 2: AI, Web & Elite Courses)

👁️ 4. Deep Learning, PyTorch & Edge Object Detection (YOLOv8)

VEDANTH (LEAD), HARIKA

Powers Subsystem C's survivor/threat classification, TensorRT edge inference, and tri-modal fusion.

RESOURCE NAME	TYPE / CREATOR	DURATION	FOCUS / STRENGTHS	LINK
YOLOv8 Full Tutorial Series	YouTube (Code With Aarohi)	4 hrs (15+ vids)	Custom dataset training, model export, real-time webcam inference	Watch Playlist ↗
PyTorch for Beginners (Deep Learning)	YouTube Series	4 hrs (10 vids)	Tensors, autograd, custom neural network modules, DataLoader	Watch Playlist ↗
Practical Deep Learning for Coders	fast.ai (Jeremy Howard)	7 Weeks (Free)	Top-down intuitive approach to computer vision and embeddings	Enroll Free ↗
CS231n: Deep Learning for Computer Vision	Stanford University	Self-Paced	Gold standard vision theory (CNN architectures, attention, detection)	Explore Course ↗
Ultralytics YOLOv8 Documentation	Ultralytics Official	Docs	Direct export commands to TensorRT FP16/INT8 engines	Read Docs ↗

🌐 5. Web Frameworks (Flask), WebSockets & Leaflet.js Mapping

SIVA (LEAD), NIKHIL

Powers Subsystem D's 3D/2D GIS Ground Control Station (GCS) and low-latency telemetry streaming.

RESOURCE NAME	TYPE / CREATOR	DURATION	FOCUS / STRENGTHS	LINK
Flask Tutorial Series	YouTube (NeuralNine)	4 hrs (10 vids)	Clean, modern Flask routing, templates, API responses, blueprints	Watch Playlist ↗
Leaflet from Basic to Advance	YouTube (GeoDev)	8 hrs (29 vids)	Custom markers, geoJSON polygons, search corridors, real-time updates	Watch Playlist ↗
Leaflet JS Tutorial 101	YouTube (MapTiler)	3 hrs (8 vids)	Rapid mapping setup, tile providers, custom overlay integration	Watch Playlist ↗
The Flask Mega-Tutorial	Miguel Grinberg	Blog / Book	The most comprehensive production guide to Flask web apps	Read Tutorial ↗

🌱 6. Git, Pytest Verification, CONOPS & Elite University Vault

HARIKA, ROHITH, ALL

Ensures repository hygiene, empirical test proofs, field rescue CONOPS narratives, and top-tier theory.

RESOURCE NAME	TYPE / CREATOR	DURATION	FOCUS / STRENGTHS	LINK
Git Mastery: From Beginner to Pro	YouTube (DevOps w/ Prashant)	4 hrs (12 vids)	Feature branching, stash, rebase, merge conflict resolution	Watch Playlist ↗
pytest Testing Framework Guide	Official Docs / RealPython	3 hrs	Test discovery, fixtures, mocking, execution timing (–durations)	Read Docs ↗
Deep Learning (MIT 6.7960)	MIT OpenCourseWare	Fall 2024	Deep neural networks, visual transformers, loss landscapes	Enroll Free ↗
NDMA Disaster Management Guidelines	Govt of India (NDMA)	Case Studies	Kedarnath flood & Wayanad landslide search corridor standard SOPs	NDMA Portal ↗

4

Subsystem Deliverables & Capability Readiness (Per Teammate)

Each teammate is free to structure their own study rhythm. The goal is satisfying these core technical capabilities and deliverables:

NIKHIL – Tech Architect & Subsystem A+B Lead

Autonomous Pace

Mission Focus: ROS2 QoS, ORCA 3D collision avoidance mathematics, PX4 offboard control, and SwarmRAFT consensus.

Recommended Reference Topics:

- ROS 2 Humble pub/sub & QoS profiles (The Construct)
- PX4 MAVLink offboard 50Hz setpoints (RIIS LLC)
- Gazebo Sim 8 SDF digital twins & sensor plugins
- ORCA 3D reciprocal velocity obstacle geometry

Target Subsystem Deliverables:

- Explain ROS2 node lifecycles & custom messages from first principles
- Derive 3D ORCA velocity obstacle cones on whiteboard
- Demonstrate 5-drone Gazebo SITL swarm with <100ms failover

VEDANTH SAI RAM – AI Perception Lead (Subsystem C)

Autonomous Pace

Mission Focus: PyTorch tensor mechanics, custom YOLOv8 training, TensorRT FP16 serialization, and DEM GPS raycasting.

Recommended Reference Topics:

- YOLOv8 custom training & export (Code With Aarohi)
- PyTorch autograd, DataLoader, and tensor dimensions
- fast.ai computer vision & attention techniques
- ROS 2 cv_bridge & image transport streaming

Target Subsystem Deliverables:

- Train lightweight survivor/threat detection model
- Derive camera raycast equation to WGS84 GPS on DEM
- Achieve TensorRT FP16 edge inference <5ms per frame

SIVA KESAVA – GCS Lead (Subsystem D)

Autonomous Pace

Mission Focus: Build a clean, responsive Flask + Leaflet.js Ground Control Station dashboard with WebSocket telemetry.

Recommended Reference Topics:

- Flask routing, templates, and REST API blueprints (NeuralNine)
- Leaflet.js custom markers, geoJSON overlays (GeoDev)
- Flask-SocketIO real-time swarm telemetry streaming
- Responsive tactical UI layout & status badge components

Target Subsystem Deliverables:

- Build live Flask server rendering active multi-UAV map
- Stream real-time coordinate packets over WebSockets
- Implement 1-Click Emergency RTL command & corridor display

HARIKA – Docs, Verification & Presentation Lead (Subsystem E)

Autonomous Pace

Mission Focus: Master pytest suites, empirical benchmarking standards (G1–G6), technical DOCS.md syncing, and pitch design.

Recommended Reference Topics:

- Python core scripting & test writing (Dave Gray)
- pytest fixtures, parameterization, and execution timing
- MIT 6.7960 / Stanford CS231n deep learning theory foundations
- Git feature branching, PR review, and commit hygiene

Target Subsystem Deliverables:

- Write comprehensive pytest unit test suites for SUTRA algorithms
- Audit all DOCS.md benchmark tables against captured live stdout
- Design master pitch presentation deck with visual system diagrams

ROHITH KUMAR – Tactical Operations Lead (Subsystem F)

Autonomous Pace

Mission Focus: Lead NDMA rescue CONOPS, field deployment SOPs, pre-flight safety checklists, and tactical storytelling.

Recommended Reference Topics:

- Disaster case studies (Kedarnath floods & Wayanad landslides)
- NDMA search and rescue protocols & geofence exclusion zones
- Markdown authoring for operational SOPs and pre-flight logs
- Operational narrative building for jury defense & presentation

Target Subsystem Deliverables:

- Draft complete NDMA CONOPS for Kedarnath & Wayanad search corridors
- Create Pre-Flight Safety & Telemetry checklists in Markdown
- Deliver 3-minute operational rescue mission narrative to the jury

5 End-to-End Swarm Pipeline, Offline Prep & Defense Strategy

SUTRA End-to-End Search & Rescue Data Pipeline (Must Master for Viva)

Every team member must be able to trace how a survivor sighting propagates across all 5 subsystems live:

1. **Optical Acquisition:** Onboard FLIR RGB/Thermal camera publishes ROS2 image message →
2. **Edge AI Detection (Subsystem C):** YOLOv8-Nano TensorRT processes frame (<5ms) → Bounding box extracted →
3. **Geolocation Raycast (Subsystem C):** Drone attitude quaternion + pinhole camera raycast onto DEM terrain → WGS84 GPS (lat/lon/alt) generated →
4. **Mesh Broadcast (Subsystem B):** Deep JSCC compressed payload broadcast over 802.11s Wi-Fi mesh to swarm peers →
5. **Distributed Consensus (Subsystem B):** SwarmRAFT leader logs survivor location into replicated swarm state machine →
6. **Collision Avoidance & Routing (Subsystem A):** ORCA 3D repels other drones from search circle while assigning rescue loiter waypoint →
7. **Tactical Display & Action (Subsystem D & F):** Telemetry pushed over WebSocket to GCS HUD → Leaflet map drops pulse marker → NDMA rescue SOP triggered.

48-Hour Offline Caching Protocol

- Pre-install full ROS 2 Humble & Gazebo Sim binaries
- Download PyTorch wheels & TensorRT FP16 models
- Cache NPM node_modules & Leaflet tile map tiles
- Keep offline HTML docs (MDN, ROS 2 docs, OpenCV)

Jury Scoring Criteria (Smart Horizon)

- **Innovation & Tech Approach (30%):** Swarm novelty
- **Hardware/SITL Feasibility (25%):** Real physics simulation
- **Swarm Fault Tolerance (20%):** Consensus & ORCA
- **NDMA Social Impact (15%):** Rescue utility
- **Code & Defense Viva (10%):** Line-by-line fluency

6 Minimum Viable Defense (MVD) — Top 7 Jury Trap Questions

Every teammate must be able to answer these core technical questions from first principles:

Q1: "Why use ORCA 3D instead of simple Artificial Potential Fields (APF)?"

Answer: APF suffers from local minima traps and oscillation. ORCA computes collision-free velocity obstacles geometrically with mathematical reciprocity guarantees, preventing deadlock among 5+ drones.

Q2: "What happens when 2 drones lose Wi-Fi mesh connectivity?"

Answer: SwarmRAFT consensus uses distributed heartbeat timers. Sub-swarms without quorum ($N/2 + 1$) transition to autonomous localized search and failsafe to last rendezvous waypoints.

Q3: "How do you calculate accurate GPS lat/long from a 2D bounding box?"

Answer: Using pinhole camera inverse projection with drone attitude quaternions (roll/pitch/yaw from VIO) to project an optical ray onto DEM terrain, converting intersection to WGS84 GPS.

Q4: "Why use Flask + Leaflet instead of heavy COTS software like QGroundControl?"

Answer: SUTRA GCS is a custom tactical multi-UAV dashboard providing real-time multi-agent swarm overlays, automated search polygon partitioning, and zero-install browser accessibility for NDMA field teams.

Q5: "How do you ensure safety under hardware or communication failure?"

Answer: Tiered failsafes: Layer 1 is 1-click Emergency RTL over WebSocket; Layer 2 is onboard PX4 geofence/battery auto-land; Layer 3 is NDMA physical exclusion zones.

Q6: "How do you prove your system achieves <5ms perception and <100ms failover?"

Answer: Strict empirical verification suites capturing live `pytest --durations` stdout, ROS 2 `topic_hz` logs, and TensorRT profiling logs rather than synthetic estimates.

Q7: "Did you generate all this code with an AI LLM during prep?"

Answer: AI served as an interactive research assistant for documentation review. The architecture, mathematical modeling, test suites, and integration nodes are deeply understood, implemented, and defended from first principles.

Event Logistics (Smart Horizon 2026)

- **Venue:** NHCE Bengaluru | Sep 3–5, 2026 (48H Offline)
- **Attendance:** All 5 team members present continuously
- **Hardware:** Laptops, power strips, USB cables, offline caches

Mandatory Submission Deliverables

- **1. Source Code:** Clean repo with tools & deps listed
- **2. Docs & Demo:** DOCS.md, test logs, SITL swarm + GCS
- **3. Deck & CONOPS:** Slides, math novelty, rescue plans

SUTRA Team Hackathon Commitment

"We commit to mastering our individual subsystems, coding with integrity, and presenting an undeniable, fully original, empirical search-and-rescue swarm system to the Smart Horizon 2026 jury."